Discrete Math W7

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Hello

I am Diego

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Group Chat



Plan

1. Feedback
2. Theory Recap
3. Exercises
4. Questions?

Feedback

Where Are We?

- Basic Logic & Proofs
- Sets & Relations
- ⇒ **Number Theory**
- Algebra
- Logic

Feedback

Theory Recap

Why Number Theory?

Mathematics is the queen of the sciences—and number theory is the queen of mathematics.

— Carl Friedrich Gauss



Why Number Theory?

I have never done anything 'useful'. No discovery of mine has made, or is likely to make, directly or indirectly, for good or ill, the least difference to the amenity of the world.

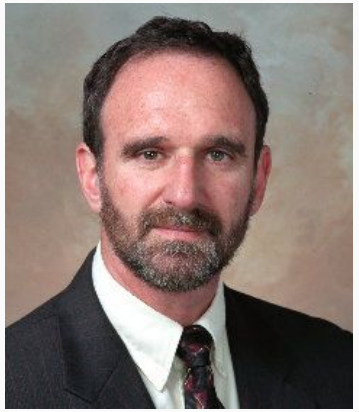
— G.H Hardy



Why Number Theory?

Whitefield Diffie and Martin Hellmann published their seminal paper on the Diffie-Hellman key exchange in 1976, building on the work of Ralph Merkle.
Since then Number Theory has had a profound impact on our digital lives and we could not have our modern internet without it.







Division

Definition 4.1

For integers a and b we say that a *divides* b, denoted $a \mid b$, if there exists an integer c such that $b = ac$. In this case a is called a *divisor* of b, and b is called a *multiple* of a. If $a \neq 0$ and a divisor c exists it is called the *quotient* when b is divided by a, and we write $c = \frac{b}{a}$ or $c = b/a$. We write $a \nmid b$ if a does not divide b.

Theorem 4.1 (Euclid)

For all integers a and $d \neq 0$ there exist unique integers q and r satisfying

$$a = dq + r \qquad \text{and} \qquad 0 \leq r < |d|.$$

Example

For $a = 23$ and $d = 7$:

$$23 = 7 \cdot 3 + 2,$$

so $q = 3$ and $r = 2$.

For a negative example, with $a = -17$ and $d = 5$:

$$-17 = 5 \cdot (-4) + 3,$$

so $q = -4$ and $r = 3$. Notice $0 \leq r < 5$.

Also, for $a = 17$ and $d = -5$:

$$17 = (-5) \cdot (-3) + 2,$$

so $q = -3$ and $r = 2$, with $0 \leq r < |-5| = 5$.

Greatest Common Divisor

Definition 4.2

For integers a and b (not both 0), an integer d is called a *greatest common divisor* of a and b if d divides both a and b and if every common divisor of a and b divides d, i.e., if

$$d \mid a \ \wedge \ d \mid b \ \wedge \ \forall c \left((c \mid a \wedge c \mid b) \rightarrow c \mid d \right).$$

Definition 4.3

For $a, b \in \mathbb{Z}$ (not both 0) one denotes the unique positive greatest common divisor by $\gcd(a, b)$ and usually calls it the greatest common divisor. If $\gcd(a, b) = 1$, then a and b are called *relatively prime* or coprime.

Example

Lemma 4.2

For any integers m, n and q, we have

$$\gcd(m, n - qm) = \gcd(m, n).$$

Example

Use Lemma 4.2 repeatedly:

$$\begin{aligned} 252 &= 2 \cdot 105 + 42, & \gcd(252, 105) &= \gcd(105, 42), \\ 105 &= 2 \cdot 42 + 21, & \gcd(105, 42) &= \gcd(42, 21), \\ 42 &= 2 \cdot 21, & \gcd(42, 21) &= \gcd(21, 0) = 21. \end{aligned}$$

Therefore $\gcd(252, 105) = 21$.

This is called Euclids Algorithm

Ideals

Definition 4.4

For $a, b \in \mathbb{Z}$, the *ideal* generated by a and b, denoted (a, b), is the set

$$(a, b) \stackrel{\text{def}}{=} \{ ua + vb \mid u, v \in \mathbb{Z} \}.$$

Similarly, the ideal generated by a single integer a is

$$(a) \stackrel{\text{def}}{=} \{ ua \mid u \in \mathbb{Z} \}.$$

Lemma 4.3

For $a, b \in \mathbb{Z}$ there exists $d \in \mathbb{Z}$ such that $(a, b) = (d)$.

Lemma 4.4

Let $a, b \in \mathbb{Z}$ (not both 0). If $(a, b) = (d)$, then d is a greatest common divisor of a and b.

Example about Ideals

Example

Consider

$$(24, 18) = \{\, 24u + 18v \mid u, v \in \mathbb{Z} \,\}.$$

Every element is a multiple of 6, since

$$24u + 18v = 6(4u + 3v).$$

Conversely,

$$6 = 24 \cdot 1 + 18 \cdot (-1),$$

so $6 \in (24, 18)$, and therefore every multiple of 6 is in $(24, 18)$. Hence

$$(24, 18) = (6).$$

Bezout's Lemma

Corollary 4.5

For $a, b \in \mathbb{Z}$ (not both 0), there exist $u, v \in \mathbb{Z}$ such that

$$\gcd(a, b) = ua + vb.$$

Euclidean Algorithms

Recall the following Lemma

Lemma 4.2

For any integers m, n and q, we have

$$\gcd(m, n - qm) = \gcd(m, n).$$

Euclidean Algorithms

Standard GCD

```
1: function GCD(a, b)
2:     if b = 0 then
3:         return a
4:     else
5:         return GCD(b, a mod b)
6:     end if
7: end function
```

This is a very good way to calculate the **gcd** of two numbers.

Euclidean Algorithms

Standard GCD

```
1: function GCD(a, b)
2:     if b = 0 then
3:         return a
4:     else
5:         return GCD(b, a mod b)
6:     end if
7: end function
```

This is a very good way to calculate the **gcd** of two numbers.

> *[The Euclidean algorithm] is the granddaddy of all algorithms, because it is the oldest nontrivial algorithm that has survived to the present day.*
>
> — Donald Knuth, Turing Award Winner

Extended Euclidean Algorithm

There is a way to modify Euclids Algorithm to also get the coefficients u and v from Lemma 4.5. You are not expected to be able to do this for the exam like it was in previous years. This information will be given to you (if necessary) in a hint.

Extended Euclidean Algorithm: Example

1. Division Steps

$$252 = \mathbf{2} \cdot 105 + \mathbf{42}$$

$$105 = \mathbf{2} \cdot 42 + \mathbf{21}$$

$$42 = \mathbf{2} \cdot 21 + 0$$

$$\Longrightarrow \gcd(252, 105) = \mathbf{21}$$

2. Back-Substitution

Isolate remainders and substitute upwards:

$$21 = 105 - 2 \cdot 42$$

$$21 = 105 - 2 \cdot (252 - 2 \cdot 105)$$

$$21 = -2 \cdot 252 + 5 \cdot 105$$

Thus, $u = -2$ and $v = 5$.

Least Common Multiple

Definition 4.5

The *least common multiple* l of two positive integers a and b, denoted $l = \mathsf{lcm}(a, b)$, is the common multiple of a and b which divides every common multiple of a and b, i.e.,

$$a \mid l \ \wedge \ b \mid l \ \wedge \ \forall m \left((a \mid m \wedge b \mid m) \rightarrow l \mid m \right).$$

Primes

Definition 4.6

A positive integer $p > 1$ is called *prime* if the only positive divisors of p are 1 and p. An integer greater than 1 that is not a prime is called *composite*.

Theorem 4.6

Every positive integer can be written uniquely (up to the order in which factors are listed) as the product of primes.

Lemma 4.7

If p is a prime which divides the product $x_1 x_2 \cdots x_n$ of some integers $x_1, \ldots, x_n$, then p divides one of them, i.e., $p \mid x_i$ for some $i \in \{1, \ldots, n\}$.

Expressing gcd and lcm

Due to the uniqueness of prime factorization we can write any integers a, b as

$$a = \prod_i p_i^{e_i} \qquad \text{and} \qquad b = \prod_i p_i^{f_i}$$

$$\gcd(a, b) = \prod_i p_i^{\min(e_i, f_i)}$$

$$\operatorname{lcm}(a, b) = \prod_i p_i^{\max(e_i, f_i)}$$

$$\gcd(a, b) \cdot \operatorname{lcm}(a, b) = ab$$

$$\min(e_i, f_i) + \max(e_i, f_i) = e_i + f_i$$

Theorem 4.8

$\sqrt{n}$ is irrational unless n is a square ($n = c^2$ for some $c \in \mathbb{Z}$).

Theorem 4.9

There are infinitely many primes.

Theorem 4.10

Gaps between primes can be arbitrarily large, i.e., for every $k \in \mathbb{N}$ there exists $n \in \mathbb{N}$ such that the set $\{n, n+1, \ldots, n+k-1\}$ contains no prime.

Example about primes

Lemma 4.12

Every composite integer n has a prime divisor $\leq \sqrt{n}$.

Example

The integer

$$221 = 13 \cdot 17$$

is composite. Since

$$\sqrt{221} \approx 14.9,$$

its prime divisor 13 satisfies $13 \leq \sqrt{221}$.

To test whether a number n is prime, it is enough to test primes up to $\sqrt{n}$. If no such prime divides n, then n must be prime.

Modular Congruences

Definition 4.8

For $a, b, m \in \mathbb{Z}$ with $m \geq 1$, we say that a is *congruent to b modulo m* if m divides $a - b$. We write $a \equiv b \pmod{m}$ or simply $a \equiv_m b$, i.e.,

$$a \equiv_m b \quad \overset{\text{def}}{\iff} \quad m \mid (a - b).$$

Lemma 4.13

For any $m \geq 1$, $\equiv_m$ is an equivalence relation on $\mathbb{Z}$.

Example 4.6 (Congruences)

We have

$$17 \equiv 2 \pmod{5}$$

because $17 - 2 = 15$ is divisible by 5. Also,

$$17 \equiv 12 \pmod{5}$$

because $R_5(17) = R_5(12) = 2$.

Basic Rules about Modular Congruences

Lemma 4.14

If $a \equiv_m b$ and $c \equiv_m d$, then

$$a + c \equiv_m b + d \qquad \text{and} \qquad ac \equiv_m bd.$$

Corollary 4.15

Let $f(x_1, \ldots, x_k)$ be a multi-variate polynomial in k variables with integer coefficients, and let $m \geq 1$. If $a_i \equiv_m b_i$ for $1 \leq i \leq k$, then

$$f(a_1, \ldots, a_k) \equiv_m f(b_1, \ldots, b_k).$$

Example

Since

$$17 \equiv 2 \pmod 5, \qquad 9 \equiv 4 \pmod 5,$$

we get

$$17 + 9 \equiv 2 + 4 = 6 \equiv 1 \pmod 5,$$

and

$$17 \cdot 9 \equiv 2 \cdot 4 = 8 \equiv 3 \pmod 5.$$

Lemma 4.16

For any $a, b, m \in \mathbb{Z}$ with $m \geq 1$,

1. $a \equiv_m R_m(a)$.
2. $a \equiv_m b \iff R_m(a) = R_m(b)$.

Corollary 4.17

Let $f(x_1, \ldots, x_k)$ be a multi-variate polynomial in k variables with integer coefficients, and let $m \geq 1$. Then

$$R_m\big(f(a_1, \ldots, a_k)\big) = R_m\big(f(R_m(a_1), \ldots, R_m(a_k))\big).$$

Corollary

For any $a, b, m \in \mathbb{Z}$ with $m \geq 1$

$$a = b \implies a \equiv_m b$$

Modular Congruence Equations

Lemma 4.18

The congruence equation

$$ax \equiv_m 1$$

has a solution $x \in \mathbb{Z}_m$ if and only if $\gcd(a, m) = 1$. The solution is unique.

Definition 4.9

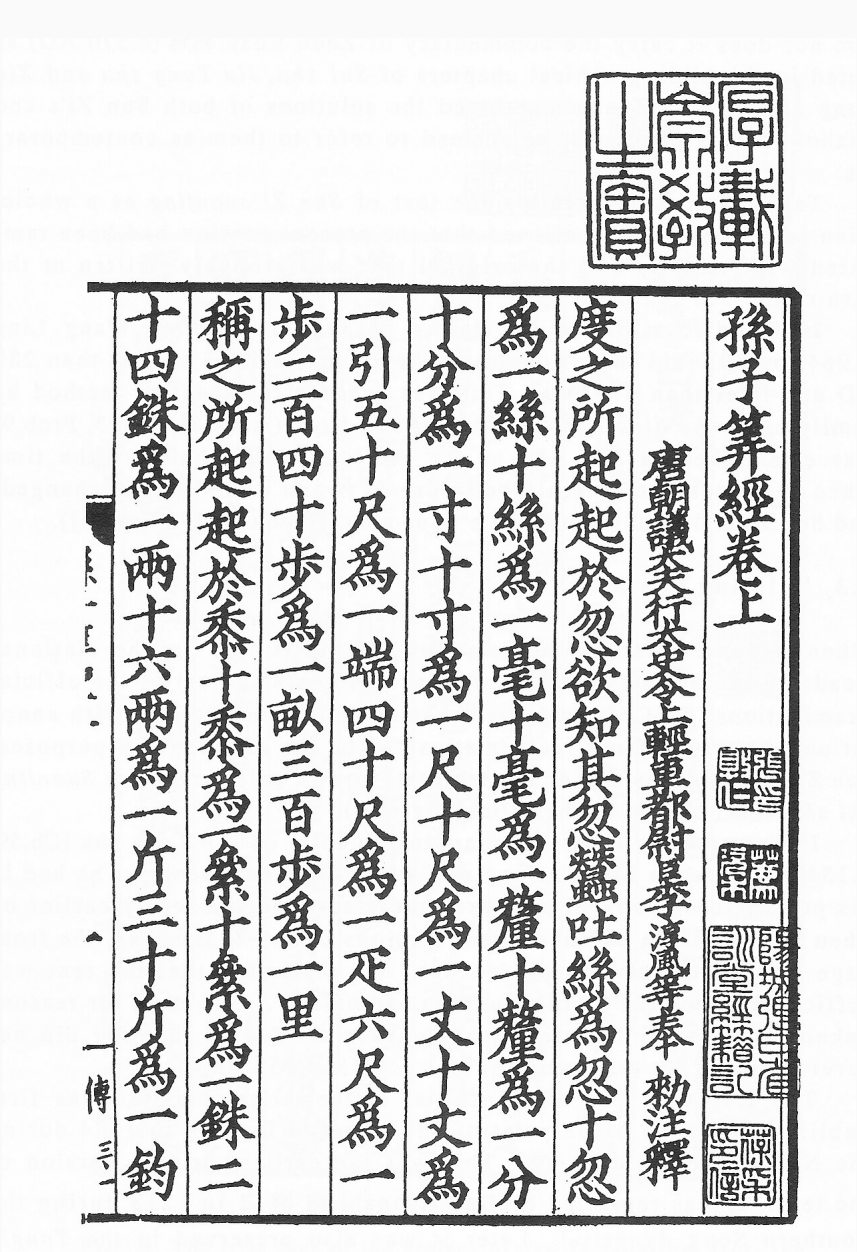
If $\gcd(a, m) = 1$, the unique solution $x \in \mathbb{Z}_m$ to the congruence equation $ax \equiv_m 1$ is called the *multiplicative inverse* of a modulo m. One also uses the notation $x \equiv_m a^{-1}$ or $x \equiv_m 1/a$.

Chinese Remainder Theorem

There are certain things whose number is unknown. If we count them by threes, we have two left over; by fives, we have three left over; and by sevens, two are left over. How many things are there?

— Sunzi, 300 – 500 CE

Chinese Remainder Theorem

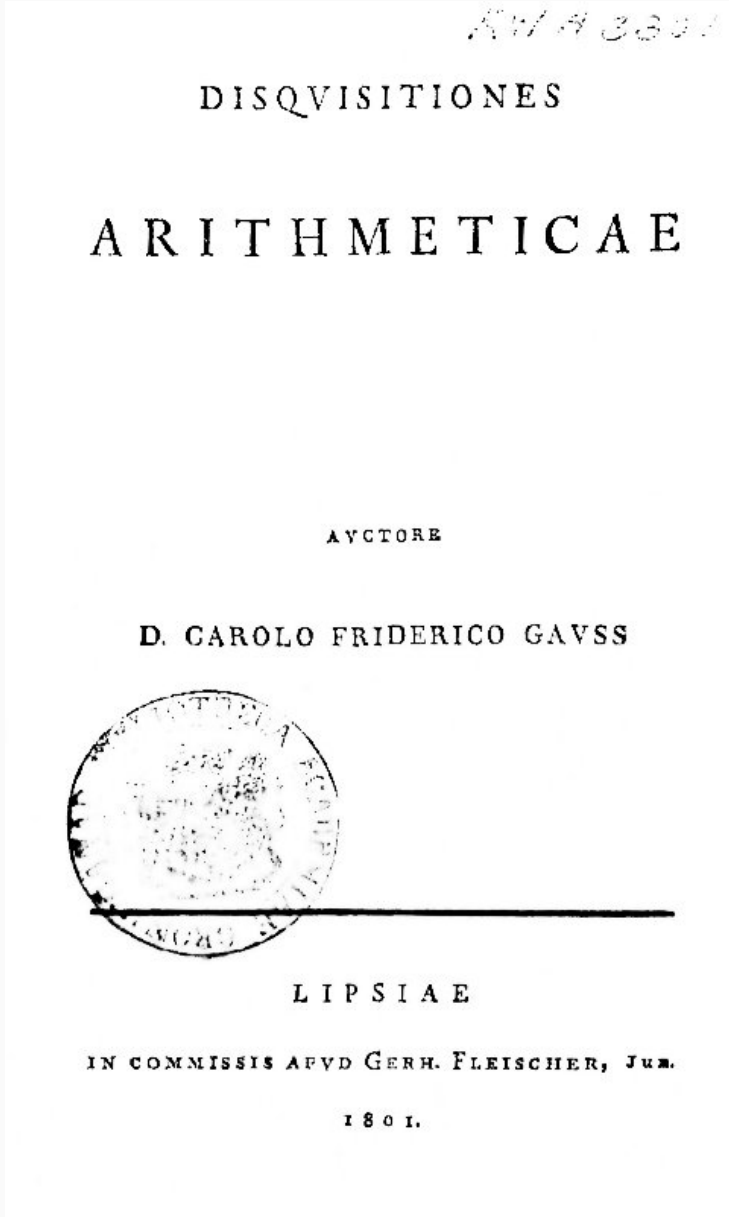


孫子算經卷上

唐朝議大夫行太史令上輕車都尉臣李淳風等奉 勅注釋

度之所起起於忽忽欲知其忽蠛吐絲爲忽十忽爲一絲十絲爲一毫十毫爲一氂十氂爲一分十分爲一寸十寸爲一尺十尺爲一丈十丈爲一引五十尺爲一端四十尺爲一疋六尺爲一步二百四十步爲一畝三百步爲一里

稱之所起起於黍十黍爲一累十累爲一銖二十四銖爲一兩十六兩爲一斤三十斤爲一鈞



DISQVISITIONES

ARITHMETICAE

AVCTORE

D. CAROLO FRIDERICO GAVSS

LIPSIAE

IN COMMISSIS APVD GERH. FLEISCHER, Jun.

1801.

Chinese Remainder Theorem

Theorem 4.19 (Chinese Remainder Theorem)

Let $m_1, m_2, \ldots, m_r$ be pairwise relatively prime integers and let $M = \prod_{i=1}^{r} m_i$. For every list $a_1, \ldots, a_r$ with $0 \leq a_i < m_i$ for $1 \leq i \leq r$, the system of congruence equations

$$\begin{aligned} x &\equiv_{m_1} a_1 \\ x &\equiv_{m_2} a_2 \\ &\vdots \\ x &\equiv_{m_r} a_r \end{aligned}$$

for x has a unique solution x satisfying $0 \leq x < M$.

CRT Example

Example

Solve

$$x \equiv 2 \pmod 3,$$
$$x \equiv 3 \pmod 5.$$

Write $x = 3 + 5t$. Modulo 3,

$$3 + 5t \equiv 2t \equiv 2 \pmod 3.$$

Since $5 \equiv 2 \pmod 3$, the inverse of 5 modulo 3 is 2. Hence $t \equiv 1 \pmod 3$, so

$$x = 3 + 5 \cdot 1 = 8.$$

Thus

$$x \equiv 8 \pmod{15}.$$

Diffie–Hellman Key Agreement

Public setup

Both parties agree publicly on:

$$p \text{ prime}, \qquad g \text{ a base}.$$

Usually p is huge, for example a 2048-bit prime.

Private values

Alice chooses a secret

$$x_A \in \{1, \ldots, p-2\}.$$

Bob chooses a secret

$$x_B \in \{1, \ldots, p-2\}.$$

Public keys

Alice computes and sends

$$y_A = R_p(g^{x_A}).$$

Bob computes and sends

$$y_B = R_p(g^{x_B}).$$

Shared secret

Alice computes

$$k_{AB} = R_p(y_B^{x_A}).$$

Bob computes

$$k_{BA} = R_p(y_A^{x_B}).$$

Why this works

The equality comes from commutativity of exponents:

$$\begin{aligned} k_{AB} &= R_p(y_B^{x_A}) \\ &= R_p\left((g^{x_B})^{x_A}\right) \\ &= R_p(g^{x_B x_A}) \\ &= R_p(g^{x_A x_B}) \\ &= R_p\left((g^{x_A})^{x_B}\right) \\ &= R_p(y_A^{x_B}) \\ &= k_{BA}. \end{aligned}$$

Thus Alice and Bob obtain the same secret number without ever sending x_A or x_B.

One-way function

It is easy to compute

$$y = R_p(g^x)$$

using repeated squaring.

Discrete logarithm problem

It is believed to bes hard to recover x from y, that is, to solve

$$g^x \equiv y \pmod{p}.$$

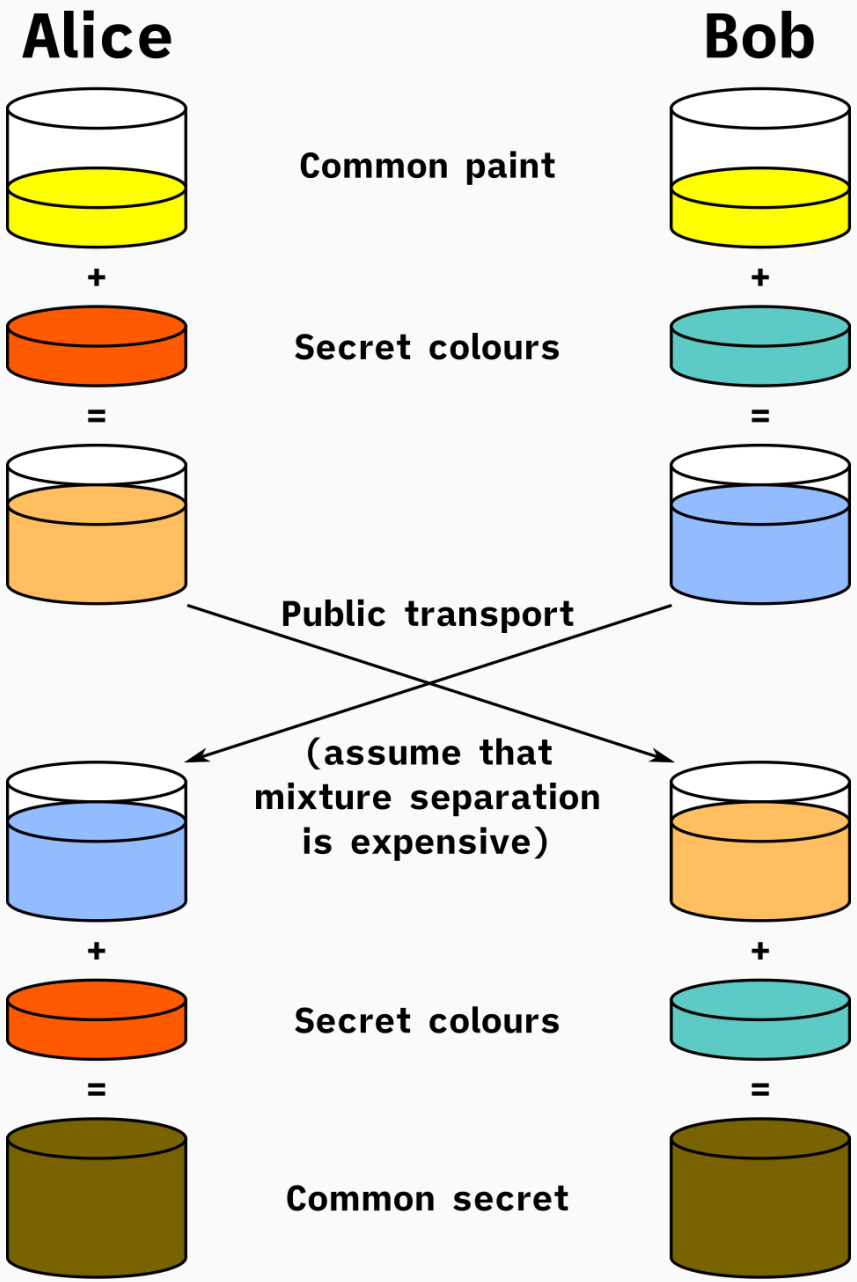
Important caveat

The public keys y_A, y_B need not be secret, but they must be authenticated. Otherwise a man-in-the-middle attacker can replace them.

Diffie–Hellman Example

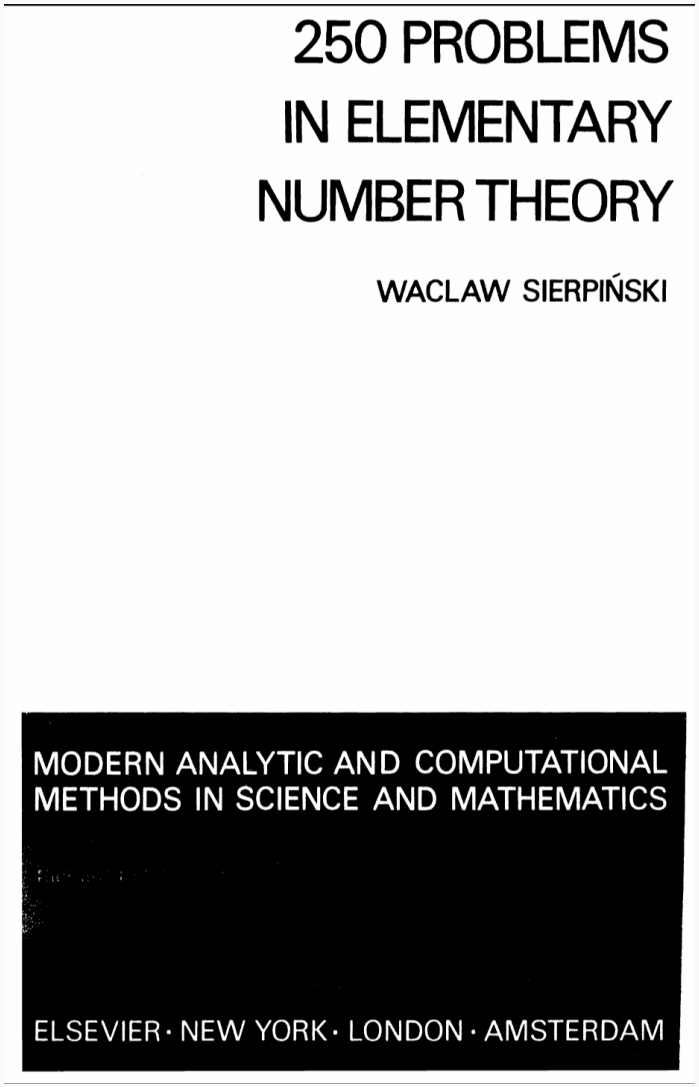
Alice	Bob
• **Secret:** $x_A = 4$	• **Secret:** $x_B = 7$
• **Computes:** $y_A = 3^4 \equiv 13 \pmod{17}$	• **Computes:** $y_B = 3^7 \equiv 11 \pmod{17}$
• **Sends** 13 to Bob.	• **Sends** 11 to Alice.
• **Shared Key:** $k = 11^4 \equiv 4 \pmod{17}$	• **Shared Key:** $k = 13^7 \equiv 4 \pmod{17}$

Observer sees: $\{3, 17, 13, 11\}$. Recovering the key requires solving the Discrete Logarithm Problem.


Alice
Bob
Common paint
+
+
Secret colours
=
=
Public transport
(assume that
mixture separation
is expensive)
+
+
Secret colours
=
=
Common secret

Exercises

250 Problems in Number Theory



Good extra exercises. Only chapters 1,2 are relevant for this course.

Divsibility

Prove that $13 \mid 2^{70} + 3^{70}$.

Solution

First notice that we have

$$2^6 = 64 \equiv -1 \pmod{13}.$$

Thus

$$2^{70} = 2^{6\cdot 11+4} = (2^6)^{11}2^4 \equiv (-1)^{11} \cdot 16 \equiv -16 \equiv -3 \pmod{13}.$$

Also,

$$3^3 = 27 \equiv 1 \pmod{13}.$$

Since $70 = 3 \cdot 23 + 1$,

$$3^{70} \equiv 3 \pmod{13}.$$

Therefore

$$2^{70} + 3^{70} \equiv -3 + 3 \equiv 0 \pmod{13},$$

so

$$13 \mid 2^{70} + 3^{70}.$$

Remainders and CRT

Prove that $11 \cdot 31 \cdot 61 \mid 20^{15} - 1$.

Solution

Since $11, 31, 61$ *are pairwise coprime, by the CRT, it is enough to prove*

$$11 \mid 20^{15} - 1, \qquad 31 \mid 20^{15} - 1, \qquad 61 \mid 20^{15} - 1.$$

Equivalently,

$$20^{15} \equiv 1$$

modulo 11*,* 31*, and* 61*.*

Solution

$$20 \equiv -2 \pmod{11}.$$

Now

$$2^5 = 32 \equiv -1 \pmod{11},$$

so

$$2^{15} = (2^5)^3 \equiv (-1)^3 = -1 \pmod{11}.$$

Therefore

$$20^{15} = (-2)^{15} = -(2^{15}) \equiv -(-1) \equiv 1 \pmod{11}.$$

Solution

$$20^2 = 400 \equiv 28 \equiv -3 \pmod{31}.$$

Hence

$$20^3 \equiv (-3) \cdot 20 = -60 \equiv 2 \pmod{31}.$$

Therefore

$$20^{15} = (20^3)^5 \equiv 2^5 = 32 \equiv 1 \pmod{31}.$$

$$20^2 = 400 \equiv 34 \pmod{61},$$

$$20^3 \equiv 34 \cdot 20 = 680 \equiv 9 \pmod{61},$$

$$20^4 \equiv 9 \cdot 20 = 180 \equiv -3 \pmod{61},$$

so

$$20^5 \equiv (-3) \cdot 20 = -60 \equiv 1 \pmod{61}.$$

Thus

$$20^{15} = (20^5)^3 \equiv 1^3 = 1 \pmod{61}.$$

CRT

Solve the following system of congruences

$$x \equiv 2 \pmod{3}$$
$$x \equiv 3 \pmod{5}$$
$$x \equiv 2 \pmod{7}$$

Solution

We use the constructive approach from the proof of the theorem. First we calculate $M = 2 \cdot 5 \cdot 7 = 105$*. We then calculate the* M_i

$$M_1 = \frac{105}{3} = 35$$
$$M_2 = \frac{105}{5} = 21$$
$$M_3 = \frac{105}{7} = 15$$

Now we solve $M_i N_i \equiv_{m_i} 1$ *for each i.*

$$M_1 N_1 \equiv_3 35N_1 \equiv_3 2N_1 \equiv_3 1 \implies N_1 = 2$$
$$M_2 N_2 \equiv_5 21N_2 \equiv_5 N_2 \equiv_5 1 \implies N_2 = 1$$
$$M_3 N_3 \equiv_7 15N_3 \equiv_7 N_3 \equiv_7 1 \implies N_3 = 1$$

Solution

Now we calculate x as

$$\begin{aligned} x &= \sum_{i=1}^{3} a_i M_i N_i \\ &= (2 \cdot 35 \cdot 2) + (3 \cdot 21 \cdot 1) + (2 \cdot 15 \cdot 1) \\ &= 140 + 63 + 30 = 233 \end{aligned}$$

We reduce this mod 105 to find the unique solution

$$233 = 2 \cdot 105 + 23 \equiv_{105} 23$$

So all solutions are of the form

$$x = 23 + 105k \quad k \in \mathbb{Z}$$

HS 26

($\star$) **Prove** that the equation

$$x^3 + y^3 = 2026$$

has no solutions $(x, y) \in \mathbb{Z} \times \mathbb{Z}$. *(4 Points)*

Solution

We consider the equation mod 7 First notice that
$2026 \equiv_7 2100 - 74 \equiv_7 -4 \equiv_7 3$

Now we calculate all remainders of x^3

x	0	1	2	3	4	5	6
x^3	0	1	1	-1	1	-1	-1

Since for no combination we have $x^3 + y^3 \equiv_7 3$*, this equation has no solution in* $(x, y) \in \mathbb{Z} \times \mathbb{Z}$

(⋆) **Prove or disprove** that there exist integers $x, y \in \mathbb{Z}$ that fulfill the following equation: *(9 Points)*

$$x^4 = y^3 + y^2 + 2$$

Corollary 5.14 (Fermats Little Theorem)

For a prime p such that $p \nmid a$ we have $a^{p-1} \equiv 1 \pmod{p}$

Solution

Consider the equation mod 5. By Fermats Little Theorem we get

$$x^4 \equiv 1 \pmod 5 \quad x \in \{1,2,3,4\} \quad \textit{and} \quad 0^4 \equiv 0 \pmod 5$$

Computing the remainders of y^3 and y^2+2 mod 5 we get

y	0	1	2	3	4
y^3	0	1	3	2	4
y^2	0	1	4	4	1
y^2+2	2	3	1	1	3
y^3+y^2+2	2	4	4	3	2

So the sets of remainders mod 5 is

$$R_5(x^4) = \{0,1\}$$
$$R_5(y^3+y^2+2) = \{2,3,4\}$$

Since these sets are disjoint, we have that $x^4 \not\equiv y^3+y^2+2 \pmod 5$. So the equation has not solutions.

Ex. 7.7.2

2. ($\star\,\star$) Prove, without using the concept of prime numbers, that for all $a, b, c \in \mathbb{N} \setminus \{0\}$ we have

$$\gcd(a, bc) \mid \gcd(a, b) \cdot \gcd(a, c).$$

Hint. For any $x, y \in \mathbb{N} \setminus \{0\}$ there exist $u, v \in \mathbb{Z}$ such that $ux + vy = \gcd(x, y)$.

Divisibility

Prove that if for some integers a, b, c we have $9 \mid a^3 + b^3 + c^3$, then at least one of the numbers a, b, c is divisible by 3.

Solution

Suppose for the sake of contradiction that none of a, b, c *is divisible by* 3*. So we have that* $a \equiv_3 b \equiv_3 \equiv_3 c \equiv \pm 1$*. Using the propertes of modular congruences we get that*

$$a^3 + b^3 + c^3 \equiv_3 \pm 1 + \pm 1 + \pm 1$$

Now one can check that all possible values of $\pm 1 + \pm 1 + \pm 1$ *are* $1, 3, -3, -1$*. None of which is divisible by* 9 *so we get that*

$$a^3 + b^3 + c^3 \not\equiv_9 0$$

which is a contradiction to our assumption. Hence at least one of a, b, c *must be divisible by* 3*.*

Fermat Numbers

Prove that for $F_n = 2^{2^n} + 1$ we have $F_n \mid 2^{F_n} - 2$ for $n = 1, 2, \ldots$

Hint: $2^n \geq 1 + n$ for $n \geq 1$

Solution

First notice that we have to show $2^{F_n} \equiv_{F_n} 2$.

By definition of F_n, *we have*

$$2^{2^n} \equiv_{F_n} -1$$

Squaring both sides yields

$$\left(2^{2^n}\right)^2 = 2^{2^{n+1}} \equiv_{F_n} (-1)^2 \equiv_{F_n} 1 \quad (*)$$

Since $n \geq 1$, *we know* $2^n \geq n+1$. *Therefore, we can write* $2^n = (n+1) + k$ *for some integer* $k \geq 0$, *which gives*

$$F_n - 1 = 2^{2^n} = 2^{(n+1)+k} = 2^k \cdot 2^{n+1} = q \cdot 2^{n+1}$$

for the integer $q = 2^k$.

Now, raising $(*)$ *to the power of* q *gives*

$$2^{F_n - 1} = 2^{q \cdot 2^{n+1}} = \left(2^{2^{n+1}}\right)^q \equiv_{F_n} 1^q \equiv_{F_n} 1$$

Solution

Multiplying both sides by 2 gives

$$2^{F_n} \equiv_{F_n} 2$$

which completes the proof.

Questions?

Material: `https://diegostrebel.com/teaching`

